

Problem-Solution Fit

Date	7 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

The canvas below maps Smart Lender's target customers, their pains, behaviors, and limitations against the proposed ML-driven solution.

1. CUSTOMER SEGMENT(S) [CS] Banks, NBFCs, and credit unions processing personal/consumer loan applications. Loan officers, credit analysts, and risk teams responsible for approval decisions.	6. CUSTOMER LIMITATIONS E.g. Budget, devices [CL] Limited budget for enterprise-grade risk/underwriting software. Limited in-house data science expertise to build and maintain ML models. Need for outputs that are explainable enough to satisfy compliance and audit requirements. Infrastructure limited to modest on-prem or cloud hardware (no large GPU clusters).	5. AVAILABLE SOLUTIONS Pros & Cons [AS] Manual underwriting — pro: human judgment; con: slow, inconsistent, doesn't scale. Rule-based scoring engines — pro: fast; con: rigid, misses complex/non-linear patterns. Generic credit bureau scores — pro: standardized; con: ignores institution-specific data. Off-the-shelf ML/credit-risk SaaS — pro: automated; con: costly, black-box, not tailored to the institution's data.
2. PROBLEMS / PAINS + its frequency [PR] Slow, manual review of applications — daily, and worse during high-volume periods. Inconsistent approval decisions across different officers. Risk of overlooking red flags in complex applicant profiles — recurring with every irregular case.	9. PROBLEM ROOT / CAUSE [RC] Absence of a unified, data-driven, real-time predictive tool that combines structured applicant data with pattern-based ML classification — forcing reliance on manual judgment or static rule-based logic that cannot adapt to nuanced or high-volume scenarios.	7. BEHAVIOR + its intensity [BE] Officers manually check income, credit history, and documents for every application — high frequency, daily occurrence. Analysts periodically batch-review portfolios of applicants — moderate to high intensity during month-end/quarter-end cycles.
3. TRIGGERS TO ACT [TR] Rising loan application volume that manual teams can't keep up with. Increasing non-performing assets (NPAs) traced back to misjudged approvals. Regulatory or audit pressure for consistent, defensible lending decisions. Competitive pressure to reduce turnaround time for applicants.	10. YOUR SOLUTION [SL] Smart Lender: a Flask-based web application that uses a trained XGBoost model to predict loan approval likelihood in real time from structured applicant data. Supports single-applicant prediction with a risk label (Low/Medium/High), Fast-Track Approval for low-risk applicants, High-Risk Detection flagging for manual review, and batch CSV evaluation for high-volume periods.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Loan officers submit applicant details and review results through the Smart Lender web portal; institutions upload batch CSV/Excel files online. OFFLINE Branch staff collect applicant documents and conduct in-person interviews before the data is entered into the system for evaluation.

	Deployable on IBM Cloud Code Engine for scalable, institution-wide use.	
4. EMOTIONS BEFORE / AFTER [EM] Before: frustration from application backlogs, anxiety about missing risky applicants, uncertainty in borderline decisions. After: confidence from data-driven, consistent decisions; relief from reduced manual workload; trust in transparent risk categorization.		